ALEXANDRE B. BARRETO

Associate Professor & Undergraduate Program Chair

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With over 30 years of cybersecurity expertise spanning academia, research, and practical application. As an Associate Professor at George Mason University, I lead courses in Cyber Security System Engineering, Secure Software Engineering, and Cyber Risk Analysis, and drive research on cybersecurity and privacy threat modeling for complex systems, including Unmanned Aircraft System Traffic Management (UTM) and crisis management systems.

As an advisor, I am advising two PhD students in Cybersecurity Engineering at George Mason, with a third joining in 2027, advising one MSc thesis at George Mason, and co-advising one MSc thesis in Brazil at Universidade de Brasília (UNB). I have already participated in 16 Academic Boards (PhD and MSc levels), with two in the first quarter of this year, following my return to Academia. I am a member of the Doctoral Supervisory Committee for several students in the CYSE and SYST Department, and I support the approval of the new Robotics PhD program.

My practical experience is extensive. I developed the Cyber-Argus framework, initially designed for ADS-B scenarios, to measure the operational impact of cyber events. This framework, published in the Journal of

Computer Networks and now foundational to the DALNIM Project (South Korean Defense), demonstrates my ability to translate research into real-world solutions. My research has focused on aviation security, including extensive work on ADS-B protocol authentication, for which I have a patent. Recent projects, such as the AFCENT counter-drone planning tool (MUSCAT), demonstrate my ability to apply advanced technologies, including HTN planning, hybrid sensor placement, and probabilistic reasoning, to address critical security challenges with civilian applications for airport and critical infrastructure protection. Ongoing research includes GPS threat analysis for GNSS procedures (UNB) and the detection of passive anomalous ADS-B attacks.

As the former Deputy Director of Research & Development at the Brazilian Air Space Control Institute (DECEA), I led multidisciplinary teams and fostered partnerships with agencies such as the FAA, ANAC, and ANATEL-BR, as well as industry leaders including Boeing and SAAB. I spearheaded the development of the DECEA certification process and managed critical projects, including the Brazilian GBAS threat model and the certification of IACIT DME and BRADAR secondary radar. My experience as a Lieutenant Colonel in the Brazilian Air Force, overseeing critical IT and security operations, has given me a deep understanding of aviation security standards and regulations.

Post-retirement, I founded a company that contributed to the approval of early Brazilian RPAS projects with ANAC and collaborated on the telecommunications and security design for the EDP-Brazilian smart grid. My leadership is characterized by strategic planning, effective communication, and a passion for enhancing cybersecurity resilience in the aviation industry, directly supporting the mission to ensure safe and secure air transport.

As the Undergraduate Program Chair for Cyber Security Engineering, I actively drive academic governance, leading comprehensive curriculum redesign initiatives that align innovative pedagogical frameworks with institutional and accreditation standards.

Education

Doctor of Science in Computer Engineering (Cyber Security)

Instituto Tecnológico de Aeronáutica (ITA), Brazil
2010-2013

Master of Science in Computer Engineering (VoIP Cryptography)

Instituto Tecnológico de Aeronáutica (ITA), Brazil
2003-2006

Master's in Business Administration - MBA (Public Management)

Universidade Federal Fluminense (UFF), Brazil
2007

Bachelor's degree in Aeronautical Science

Academia da Força Aérea (AFA), Pirassununga, Brazil
1992-1995

Languages

English — ●●●●● *Fluent*

Portuguese — ●●●●● *Fluent*

Spanish — ●●●○○ *Intermediary*

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Skills

Systems Engineering & Security Design	Pedagogical Frameworks (EML, ABET, Active Learning, TDI)	Cyber Security Architecture (DODAF, MITRE, NIST)	Decision-Making Models (Bayesian, ML, Planning, Semantic Technologies)
●●●●●	●●●●○	●●●●●	●●●●○
Excellent	*Excellent*	*Advanced*	*Excellent*

Academic Experience

Ago 2023 – Present

Associate Professor

George Mason University (GMU) – Fairfax, US.

- **Undergraduate Program Chair:** Leading departmental academic governance, driving strategic reviews, and evaluating proposals for courses, minors, professional certificates, and degree programs.
- **Curriculum Innovation Leader:** Spearheading the comprehensive CYSE BS curriculum redesign project, systematically embedding Entrepreneurial Mindset Learning (EML) techniques and active learning methodologies directly into ABET accreditation criteria.
- **International Program Architect:** Formulated and led the strategic curriculum design and academic alignment for the international double-degree undergraduate program in partnership with Konyang University (South Korea).
- **Advanced Pedagogical Training:** Completed intensive professional development in active learning and entrepreneurship education, including the DI (Teaching, Deconstructing, and Integrating) (TDI) seminar (Summer 2025) and the Entrepreneurial Mindset Learning (EML) workshop (Summer 2026), and was selected for the inaugural 2027 **EM-FUEL** (Entrepreneurial Mindset for Future University Engineering Leaders) cohort supported by KEEN and Engineering Unleashed.
- **Curriculum Architecture & Course Design (Faculty of Record):** Conceptualized and created from the ground up **CYSE 788** (Advanced Systems Engineering for Cybersecurity), a specialized doctoral-level course focusing on design, validation, and assurance frameworks for AI-enabled and autonomous systems. Additionally, redesigned and updated core departmental syllabi across foundational and graduate levels, including **CYSE 411** (Secure Software Engineering), **CYSE 550** (Cyber Security Engineering Fundamentals Tools), **CYSE 630** (Cyber Risk Analysis and Advanced Tools), and the joint **CYSE 587 / SYST 687 / CYSE 787** (Cyber Security Systems Engineering) sequence.

Feb 2021 – Dec 2022

Professor

Tecnológico de Monterrey.Mexico City (CDMX) – Mexico.

- Design and teach courses for synchronous and asynchronous models using challenge- and competence-based learning (based on the TEC21 Methodology[1]).
- Instructional Faculty and Research Member of the Tecnológico de Monterrey Cyber Hub.
- Support and advise the organization and documentation of the CTIR of TEC Cyber Hub.
- Technical Leadership and Project Management of the Cyber Crisis Simulator, a framework that emulates and simulates Advanced Persistent Threat (APT) attacks against simulated business environments, enabling managers and technical teams to work together in solving crises caused by cyber-attacks while providing training to technical and executive teams working within holistic and realistic environments.

[1] https://tec.mx/en/model-tec21

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- Develop and teach an undergraduate course based on the **Global Shared Learning**[2] methodology, where Mexicans (IoT Course) and Brazilian students from the partner university, the Instituto Tecnológico de Aeronáutica – ITA, work together (remotely) to develop IoT and drone solutions to assist the population during a weather calamity. Project website: https://github.com/kabartsjc/gsl-iot-2022.
- Professor (instructor of record) of the following undergraduate courses in Cyber Security and Network fields: **Computer and Information Security** (2021/2022), **Cryptology and Security Tools** (2021), and **Device Interconnection – Network** (2022).
- Professor of the following regular undergraduate courses in Artificial Intelligence Applied to Cyber Security: **Computational Intelligence** (2021). Students learn to find evidence and classify attacks using artificial intelligence and machine learning techniques.
- Professor of the following regular undergraduate course spanning the Software Engineering and the Cyber Security fields: **Integration of Computer Security in Networks and Software Systems** (2021). This is a block course in which various Professors alternate in teaching and supporting the students to develop a mobile and security system for an external client (an actual company). In addition, I was responsible for teaching test and validation techniques applied to the Swift Language.
- Professor of the following regular undergraduate course on the Internet of Things (IoT) field: **Implementation of Internet of Things** (2021, 2022). This is also a block course but focused on developing an IoT application. I was the course coordinator and was also responsible for introducing IoT.

Nov 2019
Feb 2021

Research Assistant Professor

George Mason University (GMU). Fairfax (VA) – USA.

- Team member of the Anytime Reasoning and Analysis for Kill-Web Negotiation and Instantiation across Domains (ARAKNID) **project**, in which CYSE and C4I & Cyber Center faculty and post-docs helped a group led by Raytheon-BBN to develop a decision-making tool for the **DARPA's Adapting Cross-domain Kill-webs program**. I designed to integrate the Hierarchical Task Network (HTN) planning rules engine with the probabilistic reasoning module. Our group also developed the latter using advanced Bayesian network technologies invented by Mason faculty (Dr. Laskey and Dr. Costa).
- Support the GMU ARAKNID Project Manager in organizing activities, implementing the SCRUM methodology, and coordinating new proposals for the project.
- Propose an innovative verification and validation process for HTN rules within the ARAKNID project, using semantic inference and formal methods.
- Design and write the new material and content for the **CYSE 587: Cyber Security Systems Engineering** course, which Dr. Paulo Costa and I taught in 2020.
- Design and write the material and content of the Automated Decision-Making Process course, taught in 2020 by Dr. Paulo Costa and me to a group of instructors and researchers from the Brazilian Army's Operations Systems Development Center in Brasília (Brazil).

2001,
2002,
and 2017

Adjunct Professor

Instituto Tecnológico de Aeronáutica (ITA). São José dos Campos (SP) – Brazil.

- Professor of the following undergraduate courses in the Software Engineering (SE) field: **CES-28 Software Engineering Fundamentals** (2017) and **CES-29 Software Engineering** (2017). Students were introduced to basic and advanced topics in SE (design patterns, automated tests, quality assurance, etc.) and developed a software solution for a sponsor (an actual company).
- Professor of the following graduate courses in Network and Security fields: **Network Management (2001, 2002) & Data Security (2001, 2002)**.

[2] https://tec.mx/en/internacionalization/global-classroom-call

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2012	**Visiting Research Scientist** **George Mason University (GMU). Fairfax (VA) – USA** • Develop the integration of the cyber domain in the **GMU/ITA C2 Collaborative Research Testbed**, which allowed the simulation and evaluation of military campaigns in cyber and physical domains (Army, Navy, and Air Force) through a distributed simulation environment located in two different sites (Brazil and U.S.A.). • Develop a framework (**Cyber-Argus**) to measure the operational impacts throw information from the cyber sensors, enabling a military commander simulates multiple course-of-actions in a holistic perspective. This framework generated several peer-reviewed papers in prestigious venues, such as the Journal of Computer Networks (in 2019). It is also the basis of the current DALNIM Project (Dependency-Based Assessment of Litigious Network Events) led by CYSE and C4I & Cyber Center Faculty.
2011	**Adjunct Professor** **Universidade Paulista (UNIP). São José dos Campos (SP) – Brazil.** • Professor of the following undergraduate courses in the Engineering Foundations field: **Numeric Computing** and **Microprocessors**. • Professor of the following graduate course: **Artificial Intelligence**.
2007, 2008, and 2011	**Adjunct Professor** **Universidade Paulista (UNIP). São José dos Campos (SP) – Brazil.** • Professor of the following undergraduate courses in the Engineering Foundations field: **Numeric Computing** and **Microprocessors**. • Professor of the following graduate course: **Artificial Intelligence**.
2009	**Adjunct Professor** **Universidade do Estado do Amazonas (UEA). Manaus (AM) – Brazil.** • Professor of the following undergraduate courses in the Network field: **Networking**. • Professor of the following undergraduate courses in the Engineering Foundations field: **Algorithms & Programming Language**, and **Programming Languages (II), Programming Language Lab**.
2004, 2005, 2008	**Adjunct Professor** **Centro Universitário Luterano de Manaus (ULBRA). Manaus (AM) – Brazil.** • Professor of the following undergraduate courses in the Network field: **Network** and **Network Protocols (I & II)**. • Professor of the following undergraduate course: **Introduction to Engineering**.
2008-2009	**Course Coordinator** **Centro Universitário Luterano de Manaus (ULBRA). Manaus (AM) – Brazil.** • Responsible for coordinating the following courses: **System Analysis** (bachelor's degree) & **Network Technologies** (college's degree). • Organize both courses, including student enrollment and financial sustainability.

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- Plan and implement the revitalization of laboratories, course syllabi, and integration with the regional industrial sector, contextualizing the courses to meet the regional demands.

2001-2002

Professor

Centro Universitário Nílton Lins. Manaus (AM) – Brazil.

- Professor of the following undergraduate courses in the Software Engineering field: **Software Engineering (I & II)**, **Multimedia Systems**, and **Advanced Topics in Software Engineering**.
- Professor of the following graduate course: **IT Systems Foundations**.

Professional Experience

Feb 2019 – Present

Founder and Director

GML Tecnologias Ltda. São José dos Campos (SP) – Brazil

- Founder of GML Tecnologias.
- Managed the consultant team responsible for designing and specifying the **EDP-Brazil smart grid** (IT/OT systems, telecommunications, security, and smart meters), generated a request-for-purpose (RFP) that supported the acquisitions, and helped EDP company in the provider's selection.
- Perform the cyber requirements elicitation, scope definition, and evaluation of security tools, frameworks, and models for the EDP-Brazil smart-grid project, using the MITRE Cyber Mission Framework during the risk analysis and architecture design.
- Managed the consultant team responsible for designing and specifying the new **EDP-Brazil Integrated Operation Center (IOC)** based on the following standards: NIST, ISO, and MITRE.
- Wrote the Test Strategy and Planning to create a new Brazilian inertial system inside a Brazilian Spatial Agency program.
- Develop a new **proposal plan (risk, safety, operational) to use a Cat 1/2 RPAS (Remotely Piloted Aircraft System) in cargo missions on offshore platforms**. The plans involve studying how to integrate the RPAS with regular helicopters and creating Unmanned Aircraft System Traffic Management (UTM) procedures and infrastructures to optimize the aircraft flows. In addition, the plan has a set of security analyses to provide protected communication links and a bunch of surveillance systems, given the criticism of the environment.

Oct 2017 – Oct 2019

Senior Project Manager

Energias Ltda. São José dos Campos (SP) – Brazil.

- Design the system architecture (software and hardware) of the project **SIAD-AERO**, which is an Unmanned Aircraft System (UAS) that recognizes faults in the power lines in a smart grid using images sensed by different sensors (Visible, IR, and UV) and machine learning (deep learning), inside an R&D grant of the Brazilian Electric Energy Agency (ANEEL).
- Manage the technical and business requirements within the SIAD-AERO sponsor (EDP-Brazil).
- Manage the software team responsible for developing inspection system features, such as the planning and machine learning algorithms.

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Oct 2013 – Oct 2017

Deputy Director of Research & Development

Instituto de Controle do Espaço Aéreo (ICEA). São José dos Campos (SP) – Brazil.

- Wrote the main research directions to the Aerospace Control Department (DECEA / Brazilian Air Force - BAF), conducting partnerships with Government Agencies (Secretary of Brazilian Civil Aviation - SAC), Companies (Boeing, SAAB, and others), and Brazilian and international Universities (GMU, ITA, USP, UNB, and others).
- Managed the **two first Brazilian certification processes of ATC systems** (Bradar's Radar and IACIT DME) and **the international bi-lateral certification process** of the Ground-Based Augmentation System (GBAS) in the equatorial region (US/FAA. & DECEA).
- Wrote a semantic cyber-security architecture framework (**Cyber-Argus**) that maps cyber assets with the business processes, aiming that cyber-threats impact (MITRE ATT&CK) can be understood and measured using KPI, enabling defensive actions could be created automatically.
- Member of an international team developed a new security protocol and system architecture that provides integrity and confidentiality of messages exchanged between aircraft using the new aircraft surveillance system (**ADSBSec - US Patent No.11022696**).
- Advised the Brazilian Air Force General Staff in studies related to creating a C4I & Cyber Research Center and other cybersecurity initiatives.
- Responsible for designing the security and network architecture to provide surveillance and weather forecasts to the Air Defense during the Brazilian World Cup, accountable for managing the remote resources (deployed in 10 different locations) and the crisis management room to operate the system.

Feb 2008 – Dec 2009

IT Manager

IV Air Defense and ATC Center – CINDACTA IV (BAF). Manaus (AM) – Brazil.

- Responsible for designing the new IT and security architecture of the **Amazon Surveillance System (SIVAM)**, which is a complex surveillance system used for monitoring Amazônia Legal ("legal Amazon area"), is an extensive network (>10.000 users and > 40 sites) that use links provided by satellite and high-speed backbones. This area includes the Brazilian rainforest to curb the trafficking of illegal narcotics and curb unlawful logging or burning of the forest. The system uses fixed and mobile ground radar and airborne surveillance using the Embraer ERJ 145.
- Managed the implementation of the routing protocol to OSPF in the Amazon region.
- Wrote the security policies and implemented the firewall and IDS architecture using open-source technologies.
- Implemented the SCRUM methodology in migrating the legacy systems to open-source technologies (Desktop, Servers, Database, and web applications).
- Wrote a new protocol that provides integrity, authentication, and confidentiality between VoIP communications (**IETF MIKEY-DHHMAC-SAS**) without the requirement of PKI Infrastructure.
- Developed a C2 system (voice and data) that provides secure mobile communication between C2 forces using VoIP technology and the newest authentication protocol, named MIKEY-DHHMAC-SAS, enabling military operations can be performed using public links.
- Managed the NOC during military operations.
- Organized the first Security Office concept in the Airspace Control Department (DECEA).

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Research Areas and Keywords

IEEE Thesaurus keywords: Agentic AI; Threat modeling; Trusted computing; Air traffic control; Global navigation satellite system; Autonomous aerial vehicles; Multi-agent systems; Bayesian networks; Command and Control

Primary areas: Security and trust assurance of agentic AI and LLM-based systems; Aviation cybersecurity, including GNSS jamming and spoofing, ADS-B security, and UAS Traffic Management; Cyber mission impact assessment

Secondary areas: Model-based systems engineering for cybersecurity; Cyber-physical and industrial control systems security; Engineering education and Entrepreneurial Mindset Learning

Methods: Bayesian networks and probabilistic reasoning; Multi-agent reinforcement learning; Semantic technologies and ontologies; MBSE with SysML; Threat modeling; Risk Analysis; Simulation-based testbeds

Application domains: Air traffic management; Unmanned aircraft systems and UTM; Satellite navigation; Smart grid and distributed energy resources; Command and control systems

Publication List

1. Zunayeed Bin Zahir, Antonio F. Crespo, and Alexandre D. Barreto. **A Scalable Multi-Layer GNSS Cybersecurity Testbed for Analyzing and Quantifying System-Level Impacts of Jamming and Spoofing in Aviation**. In: 2026 IEEE/AIAA 45th Digital Avionics Systems Conference (DASC), Orlando, FL, USA, 2026. Best of Session Award.
2. A. Barreto, S. Matsumoto, J. Valverde-Rebaza, C. Ataide, and P. Costa. **Toward a Decision-Assurance Layer for AI-Assisted Flight Planning in Air Traffic Management**. arXiv preprint arXiv:2609.13552, 2026.
3. L. B. de Lima, J. C. Valverde-Rebaza, A. de B. Barreto, and Y. Iano. **AETHON: Deterministic Pre-Inference Defense for LLM Prompt Injection**. In: Proceedings of the 12th Brazilian Technology Symposium (BTSym'26): Emerging Trends and Challenges in Technology. Springer Nature, 2026.
4. Brooke Prigg, Ali K. Raz, and Alexandre D. Barreto. **Cybersecurity Threat Analysis of a Free-Space Optical Communication System Through Model-Based Systems Engineering**, AIAA 2026-2496. AIAA SCITECH 2026 Forum. January 2026.
5. M. I. Raja, B. N. Prigg, S. Akram, M. Dhanoa, A. Islam and A. D. B. Barreto, **A Comprehensive Security and Privacy Analysis of the Uncrewed Aircraft System Traffic Management (UTM) - A Cyber Security System Perspective**, 2025 AIAA DATC/IEEE 44th Digital Avionics Systems Conference (DASC), Montreal, QC, Canada, 2025, pp. 1-10, doi: 10.1109/DASC66011.2025.11257358.
6. Davila-Andino, A., Barreto, and Huang, E. **A SysML Metamodeling Approach to Embedding STRIDE Threat Analysis for Cyber-Physical Systems** [Manuscript submitted for publication]. Conference on Systems Engineering Research, 2025.
7. Albuquerque, P. K. d., Santos, W. G. d., Costa, P., & Barreto, A. (2024). **Integrating Cosmic Microwave Background Readings with Celestial Navigation to Enhance Deep Space Navigation.** Sensors, 24(11), 3600. https://doi.org/10.3390/s24113600
8. S. Matsumoto, A. Barreto, P. C. G. Costa, B. Benyo, M. Atighetchi, and D. Javoršek. **Dynamic Explanation of Bayesian Networks with Abductive Bayes Factor, Qualitative Propagation, and Entropy-Based Qualitative Explanation**. 2021 IEEE 24th International Conference on Information Fusion (FUSION), 2021, pp. 1-9, doi: 10.23919/FUSION49465.2021.9626961.
9. Crespo, A. et al. **Learning framework for carbon emissions predictions incorporating a RReliefF driven features selection and an iterative neural network architecture improvement.** SN Applied Sciences, v. 3, n. 4, p. 1-18, 2021.

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10. Leite, D. V., Weigang, L. , Barreto, A. B. and Crespo, A. M. F. **Computational Solution to Prevent Aeronautics Accidents Cause by Wake Turbulence Using Machine Learning.** IECON 2020 The 46th Annual Conference of the IEEE Industrial Electronics Society, Singapore, Singapore, 2020, pp. 124-129, DOI: 10.1109/IECON43393.2020.9255032.
11. Dominice, A., et al. **SIAD-AERO: A New Methodology for the Inspection of Energy Assets.** In: Proceedings of the 21st International Symposium on High Voltage Engineering. Springer International Publishing, 2019. p. 243-252.
12. Barreto, A. B.; Costa, PCG. **Cyber-ARGUS-A mission assurance framework.** Journal of Network and Computer Applications, v. 133, p. 86-108, 2019.
13. Barreto, A. B. et al. **Smart-Grid Assets Inspections-Enabling the Smart Cities Infrastructure.** In: 2018 International Conference on Computational Science and Computational Intelligence (CSCI). IEEE, 2018. p. 531-536.
14. Bonomo, I. Ramos, I. Monteiro, L., Borges, V. Weigang, L. Basseto, C., Barreto, A.B. **Development of SWIM Registry for Air Traffic Management with the Blockchain Support.** In: 2018 IEEE 21st International Conference on Intelligent Transportation Systems (ITSC), 2018, Maui. 2018 IEEE 21st International Conference on Intelligent Transportation Systems (ITSC), 2018 (forthcoming).
15. Kacem, Thabet; Wijesekera, Duminda; Costa, Paulo; Barreto, Alexandre B. **Extending ADS-B for Mixed Urban Air Traffic.** In: 2018 IEEE/AIAA 37th Digital Avionics Systems Conference (DASC), 2018, London. 2018 2018 IEEE/AIAA 37th Digital Avionics Systems Conference (DASC).
16. Kacem, Thabet; Wijesekera, Duminda; Costa, Paulo; Barreto, Alexandre B. **Secure ADS-B framework ADS-BSec.** In: 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC), 2016, Rio de Janeiro. 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC), 2016. p. 2681.
17. Monteiro, Marcio; Sarmento, Thalysson; Barreto, Alexandre; Costa, Paulo; Hieb, Michael. **An integrated mission and cyber simulation for Air Traffic Control.** In: 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC), 2016, Rio de Janeiro. 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC). p. 2687.
18. Monteiro, M.; Sarmento, T.; Barreto, Alexandre; Costa, P.C. **A Holistic Approach to Evaluate Cyber Threat.** In: STIDS 2016 - Semantic Technology for Intelligence, Defense, and Security, 2016, Fairfax. Proceedings of the Eleventh Conference on Semantic Technology for Intelligence, Defense, and Security, 2016. p. 64-68.
19. Monteiro, M.; Barreto, A.; Kacem, T.; Carvalho, J.; Wijesekera, D.; Costa, P. Detecting malicious ADS-B broadcasts using wide-area **multilateration.** In: 2015 IEEE/AIAA 34th Digital Avionics Systems Conference (DASC), 2015, Prague. 2015 IEEE/AIAA 34th Digital Avionics Systems Conference (DASC). p. 4A3-1.
20. Monteiro, M.; Barreto, A.; Kacem, T.; Wijesekera, D.; Costa, P.C. **Detecting malicious ADS-B transmitters using a low-bandwidth sensor network.** In: 2015 18th International Conference on Information Fusion (Fusion), 2015, Washington. 2015 18th International Conference on Information Fusion (Fusion), 2015. p. 1696-1701.
21. Kacem, Thabet; Wijesekera, Duminda; Costa, Paulo; Carvalho, Jeronymo; Monteiro, Marcio; Barreto, Alexandre**. Key distribution mechanism in secure ADS-B networks.** In: 2015 Integrated Communication, Navigation, and Surveillance Conference (ICNS), 2015, Herdon. 2015 Integrated Communication, Navigation and Surveillance Conference (ICNS), 2015. p. P3-1-P3-13.
22. Monteiro, G. S.; Barreto, A.; Oliveira, J. M. P.; Costa, I. C. **Uma abordagem para inclusão de segurança na descrição semântica de serviços Web no contexto do SISCEAB.** In: XVII Simpósio de aplicações operacionais em áreas de defesa, 2015, São José dos Campos. XVII Simpósio de aplicações operacionais em áreas de defesa, 2015. p. 28-33.
23. Kacem, Thabet; Carvalho, Jeronymo; Wijesekera, Duminda; Costa, Paulo; Monteiro, Márcio; Barreto, Alexandre**. Risk-adaptive Engine for Secure ADS-B Broadcasts.** In: SAE 2015 AeroTech Congress & Exhibition, 2015.
24. Kacem, Thabet; Wijesekera, Duminda; Costa, Paulo; Carvalho, Jeronymo; Monteiro, Marcio; Barreto, Alexandre. **Secure ADS-B design & evaluation.** In: 2015 IEEE International Conference on Vehicular Electronics and Safety (ICVES), 2015, Yokohama. 2015 IEEE International Conference on Vehicular Electronics and Safety (ICVES), 2015. p. 213.

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26. Costa, I. C.; Monteiro, G. S.; De Barros Barreto, A.; Oliveira, J. M. **Uma Proposta de Arquitetura baseada no SWIM para a implementação do Airport-CDM.** In: Simpósio de Aplicações Operacionais em Áreas de Defesa, 2014, São José dos Campos. Simpósio de Aplicações Operacionais em Áreas de Defesa, 2014. p. 119-124.
27. Kacem, T.; Wijesekera, D.; Costa, P.C.; Barreto, A.B. **Security Requirements Analysis of ADS-B Networks.** In: STIDS 2014 Semantic Technology for Intelligence, Defense, and Security, 2014, Fairfax. STIDS 2014 Semantic Technology for Intelligence, Defense, and Security, 2014. v. 1304. p. 40-47.
28. Costa, I. C.; Barreto, A.B.; Oliveira, J. M. **Service selection approach for swim consumer application based on semantic annotations in SOAML architecture.** In: Digital Avionics Systems Conference (DASC), 2014, Colorado Springs. Digital Avionics Systems Conference (DASC), 2014. p. 8B2-1-8B2-9.
29. Barreto, A.B.; Costa, P. C. G.; Yano, E. T. **Using a Semantic Approach to Cyber Impact Assessment.** In: STIDS 2013 - Semantic Technologies for Intelligence Defense and Security, 2013, 2013, Fairfax. Proceedings of the Eighth Conference on Semantic Technologies for Intelligence, Defense, and Security, 2013. v. 1097. p. 101-108.
30. Machado, A. F. A.; Barreto, A.B.; Yano, E. T. **Architecture for Cyber Defense Simulator in Military Applications.** In: 18th International Command and Control Research and Technology Symposium, 2013, Alexandria. 18th International Command and Control Research and Technology Symposium, 2013.
31. Pullen, J.M.; Ababneh, M.; Nicklas, L.; Connor, M.; Barreto, A.B. **An Open Source MSDL/C-BML Interface to VRForces.** In: 2012 Fall Simulation Interoperability Workshop, 2012, Orlando. A DoD MODELING & SIMULATION COORDINATION OFFICE PUBLICATION, 2012. v. WINTER. p. 28-35.
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33. Barreto, A.B.; Hieb, M.; Yano, E.T. **Developing a Complex Simulation Environment for Evaluating Cyber Attacks.** In: The Interservice/Industry Training, Simulation & Education Conference (I/ITSEC), 2012, Orlando. The Power of Innovation Enabling the Global Force, 2012. v. 2012.
34. Baruffaldi, A.L.M; Barreto, A.B.; Oliveira, J. M. P.; Marques, H.C. **An Automated Data Fusion Process for an Air Defense Scenario.** In: 16th ICCRTS, 2011, Quebec. Collective C2 in Multinational Civil-Military Operations, 2011.
35. Barreto, A.B.; Yano, E.T. **A Influência de Arquiteturas Abertas em Sistemas de Defesa para Obtenção de Requisitos de Dependabilidade.** In: XIII SIGE, 2011, São José dos Campos. Simpósio de Aplicações Operacionais em Áreas de Defesa, 2011.
36. Marques, H.C.; Oliveira, J.M.P.; Barreto, A.B.; Costa, P.C.; Hieb, M**. Distributed Simulation as a Planning Support Tool for Relief Operations.** In: XIII SIGE, 2011, São José dos Campos. XIII SIGE, 2011.
37. Barreto, A.B.; Yano, E.T. **Using a P2P architecture for voice and radar transportation in critical Command and Control systems.** In: MILCOM 2010 2010 IEEE Military Communications Conference, 2010, San Jose. 2010 - MILCOM 2010 MILITARY COMMUNICATIONS CONFERENCE. p. 1169.
38. Barreto, A.B.; Goncalves, R. L.; Faleiros, A. C. **MIKEY-DHHMAC-SAS - The New MIKEY Transportation Mode.** In: WORLDCOMP'07, 2007, 2007, Las Vegas, NA. The 2007 International Conference on Multimedia Systems and Applications (MSA'07), 2007. v. 1. p. 117-123.

Patent

Kacem, T., Wijesekera, D., Da Costa, P. C. G., & de Barros Barreto, A. (2021). **ADS-Bsec: a holistic framework to secure ADS-B.** US Patent No. 11,022,696. Washington, DC: US Patent and Trademark Office.

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Pending Proposals

SaTC 2.0: RES: Distributed and Trustworthy Multi-Agent Resilience to GNSS Interference in Safety-Critical Cyber-Physical Systems (COMPASS): National Science Foundation, 3 years, Role: PI, in review, $1,200,000.

Cyber Resilience for Air Traffic Management Systems (CR-ATM): Department of Airspace Control (DECEA), Brazil, Role: PI, in review, $1,200,000.

CRAFTERS LAB: U.S. Army, Role: Co-PI, in review, $1,000,000.

Academic Advisory

Zunayeed Bin Zahir (2025) – in development

Title: A distributed reinforcement learning algorithm to evaluate the impact of jamming and spoofing in the GNSS navigation system.
Level: Doctoral Program (Cyber Security) - advisor.
School: George Mason University

Shafaque Shahid Qureshi (2026) – in development

Title: Adversarial degradation of distributed multi-agent missions and agent trust in UAS.
Level: Doctoral Program (Cyber Security) – advisor.
School: George Mason University

Dip Gourab Isaac Gomes (2027) – incoming

Title: Adversarial degradation of distributed multi-agent missions and agent trust in aviation protocols.
Level: Doctoral Program (Cyber Security) – advisor.
School: George Mason University

Md Faisal Ahmed (2026)

Title: Belief-Driven Agentic Web Reconnaissance.
Level: Master's Program (Cyber Security) – co-advisor.
School: George Mason University.

Carlos Eduardo Paes (2025) – in development

Title: Standards and Regulation to combat the illegal interference in GNSS communication – a technical perspective.
Level: Master's Program (Electrical Engineering) – co-advisor.
School: Universidade de Brasília (UNB) – Brazil.

Matt Manganello (2026) – in development, expected 2027

Title: A physics-informed cyber-physical security framework for DER aggregation, grid-edge and DER security.
Level: Master's Program (Cyber Security) – advisor.
School: George Mason University

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Gerson Monteiro Siqueira (2016)

Title: An Automatized Method to Insert Annotations, Measure, and Classification Vulnerabilities into Semantic Web Services.
Level: Master's Program (Electronic and Computer Engineering) – co-advisor.
School: Instituto Tecnológico de Aeronáutica (ITA) - Brazil

Academic Boards Summary

Type	2025	2021	2020	2018	2017	2016	2015	2014
Ph.D. Thesis		01			03			
Doctoral Qualification Board	03		01				01	01
MSc Dissertation	02			02	02	02		01

Other Evaluation Boards

Aviation Knowledge and Innovation Award: Member of the Judging Committee of the Aviation Knowledge and Innovation Award granted by the Brazilian Aviation Secretary (SAC) – 2016.

InovANAC Safety Aerodromes Award: Member of the Judging Committee of the InovANAC Safety Aerodromes Award granted by the National Civil Aviation Agency (ANAC) – 2018.

Canada Research Chairs Program, NSERC: External Reviewer for the Tri-agency Institutional Programs Secretariat, Government of Canada, evaluating a Tier 2 Canada Research Chair renewal nomination in Cyber Security and Threat Intelligence, 2024.

Academic Filiations

IEEE Computer Society **Senior Member** #92401837

C4I & Cyber Affiliate Faculty at George Mason University (Fairfax – VA - US)

Academic Awards

1st. Place Professional Paper Entitled: Key distribution mechanism in secure ADS-B networks. In: 2015 Integrated Communication, Navigation, and Surveillance Conference (ICNS).

Best Cyber Security Paper Entitled: Key distribution mechanism in secure ADS-B networks. In: 2015 Integrated Communication, Navigation, and Surveillance Conference (ICNS).

Best of Session Award: A Scalable Multi-Layer GNSS Cybersecurity Testbed for Analyzing and Quantifying System-Level Impacts of Jamming and Spoofing in Aviation. In: 2026 IEEE/AIAA 45th Digital Avionics Systems Conference (DASC), session GNSS Threat Detection, Integrity & Resilience Testing.

Journal Reviewer

Elsevier Journal of Network and Computer Applications

Elsevier International Journal of Critical Infrastructure Protection

Elsevier Transportation Engineering

Military Awards

Military Medal Silver granted by the Brazilian Air Force (2011).

Military Bronze Medal granted by the Brazilian Air Force (2018).

Medal of Merit Santos-Dumont granted by the Brazilian Air Force (2009).

Invited Speakers

Cyber-Argus Framework – Measuring Mission Impact through Cyber Data. Modeling and Simulation Support for Cyber Defense Workshop, NATO Modeling and Simulation Group (MSG-121), April 2013, California (USA).

Research Projects

GMU/ITA C2 Collaborative Research Testbed 2010 - 2013	The GMU/ITA C2 Testbed is a partnership between George Mason University (GMU) and the Aeronautics Institute of Technology – Brazil (ITA) that aims to create an environment to support C2 research by providing a Modeling and Simulation Environment for C2 Planning, Security Issues, and Cyber Warfare. This environment uses several COTS (commercial off-the-shelf) tools and Open Standards that provide a rapid prototyping and modeling environment for C2 missions. The Testbed supported different Ph.D. projects, such as demonstrating operational concepts to real operators from the Brazilian Air Force.
Cyber-Argus 2013 - 2015	Cyber-ARGUS is a Command and Control (C2) support system comprised of tools that provide a coherent and consistent mapping between the two domains. Relevant information about the nodes of a cyberinfrastructure supporting an operation is stored in a knowledge base and then used to build a Bayesian Network that provides impact assessment. This environment was integrated with the Brazilian Air Force's real Air Traffic Management (ATM) system. It enabled the evaluation of the operational impacts in real air traffic control missions. It used an internal budget from the Brazilian Air Force, around a million dollars, and engaged 2 Ph. D.S, 2 M.SCs, and 2 Engineers.
SIAD-AERO	The SIAD-AERO project contemplates using drones for autonomous/assisted inspection, thus replacing human and manual activity. The project involves two categories of drones: multirotor

2017 - 2019	(used for vertical and horizontal inspections at small distances with a high index of visual accuracy) and fixed-wing (used for high-speed, long-distance flights). Both are equipped with high-resolution cameras, infrared sensors, dual GPS systems, a flight-planning system used in aeronautics, an image-recognition system linked to artificial intelligence, parachutes, and a mobile station for assisted operation at ground level. The Brazilian Electric Energy Agency (ANEEL) funded the five-million-dollar project.
ARAKNID 2017 - 2019	The Anytime Reasoning and Analysis for Kill-Web Negotiation and Instantiation across Domains (ARAKNID) project was developed as part of the DARPA Adapting Cross-Domain Kill-Webs program, which pushes the forefront of integrated deterrence by connecting real-world air domain sensors and creating, scoring, and visualizing multiple courses of action in real-time—faster than air battle managers can manually create a single course of action [COA]. The C4I & Cyber Center supports Raytheon-BBN technologies by providing state-of-the-art research in Applied Explainable AI and Probabilistic Semantic Technologies.
MUSCAT 2024 – NOW	George Mason University C5I Center research project, where the research team is developing a drone operations analysis testbed, Mason's UAV Systems and Cyber Analysis Testbed (MUSCAT), sponsored by the Commonwealth. This testbed combines JDL Data Fusion model-theoretic foundations with systems engineering principles applied to information systems for UAV corridor analysis, guiding multi-sensor acquisition decisions. This testbed combines JDL Data Fusion model-theoretic foundations with systems engineering principles applied to information systems for UAV corridor analysis, guiding multi-sensor acquisition decisions.

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